

The missing $\lambda > 2$ injectivity range is answered

Literature identification for arXiv:2201.07837

Automated mathematics run `fa_banach_001`

Lane 14, model GPT5.6

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Status: `literature_already_answered`.

Source question

Kania and Lewicki's companion paper *A forgotten theorem of Pełczyński: (λ^+) -injective spaces need not be λ -injective - the case $\lambda \in (1, 2]$* (arXiv:2201.07837) proves the result for $1 < \lambda \leq 2$. In the Introduction, immediately after Theorem A, it states:

Furthermore, the complementary case of $\lambda \in (2, \infty)$ remains a mystery to us.

Thus the exact missing assertion is the existence, for every $\lambda > 2$, of a (λ^+) -injective Banach space which is not λ -injective.

Answering theorem

The same authors' later paper *(λ^+) -injective Banach spaces* (arXiv:2603.09710, 2026) explicitly says in its abstract and Introduction that it resolves the complementary range left open in the companion paper. Its Theorem A states:

For every $\lambda > 1$ there exists a Banach space that is (λ^+) -injective but not λ -injective.

This contains the entire missing interval $(2, \infty)$, so it is an exact affirmative answer.

Identification

The answering paper introduces the zero-sum subspace

$$\Sigma_N(Y) = \left\{ (y_1, \dots, y_N) \in Y_\infty^N : \sum_{j=1}^N y_j = 0 \right\}.$$

Its multiplication lemma shows that the relative projection constant is multiplied by

$$\mu_N = 2 - \frac{2}{N},$$

and that failure of attainment is preserved. Iterating this construction reduces any prescribed $\lambda > 2$ to a base constant in $(1, 2]$ provided by arXiv:2201.07837.

Scope and provenance

The supporting authors explicitly know they are completing their earlier open range, so the appropriate status is `literature_already_answered`, not a new proof produced by this run. Together the two papers establish Pełczyński's claimed theorem for all $\lambda > 1$. The complex endpoint $\lambda = 1$ remains outside this answer.

References

- [1] T. Kania and G. Lewicki, *A forgotten theorem of Pełczyński: $(\lambda+)$ -injective spaces need not be λ -injective - the case $\lambda \in (1, 2]$* , Studia Math. (2023), arXiv:2201.07837.
- [2] T. Kania and G. Lewicki, *(λ^+) -injective Banach spaces*, arXiv:2603.09710 (2026).